

CS 307 Design Document

Aegis Project Management System



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Design Document

Purpose:

Current task management tools like are overloaded with features that are overwhelming and have a steep learning curve, especially for small teams and individual users who need a straightforward way to manage tasks. The Aegis project management system will focus on simplicity, prioritizing core features like task creation, assignment, and progress tracking without unnecessary complexity. Its user interface will be simple and intuitive, making it accessible for those unfamiliar with using similar systems.

The purpose of this project is to develop a task-management website that can be used by organizations to manage projects by enabling the delegation of specific tasks to people, managing associated files, keeping track of the progress on each, and enabling organization-level communication between groups of users. Compared to other existing applications, it is designed to have an easy-to-use interface for assigning tasks and managing the tasks assigned to a user. We hope that this application will make it easier for organizations unfamiliar with similar software to utilize and that it helps make the advantages of task-delegation software more broadly accessible for a greater variety of people.

Requirements

Functional:

All users:

1. As an account holder, I would like to create an account for Aegis.
2. As an account holder, I would like to log into my Aegis account from different devices.
3. As an account holder, I would like my login information to be stored in a secure fashion.
4. As an account holder, I would like to be able to reset my password with an email.
5. As an account holder, I would like to be able to opt into using two-factor authentication and receive a code in an email to log into my account.
6. As an account holder, I would like to sort my tasks differently based on urgency, assigner, and project in the forms of a to-do list, kanban board or calendar.
7. As an account holder, I want to switch between different views (to-do list, kanban board, calendar) to organize my tasks according to my preference
8. As an account holder, I want to filter tasks by urgency levels (e.g., high, medium, low) to prioritize my work.
9. As an account holder, I want to sort tasks by the person who assigned them to me, so I can manage expectations and communication.
10. As an account holder, I want to group tasks by project to see all related tasks in one place.
11. As an account holder, I want to search my list of tasks for those containing specific phrases or that have specific urgency levels.
12. As an account holder, I would like to view project progress reports so that I can assess team performance.
13. As an account holder, I would like to be able to message other account holders, through direct messages or in larger group chats (specific to a task or a project).
14. As an account holder, I want to initiate direct messages with other users to discuss specific issues privately.
15. As an account holder, I want to create group chats for specific tasks to facilitate team collaboration.
16. As an account holder, I want to create project-specific chat rooms to discuss broader project goals and updates.
17. As an account holder, I want to receive notifications for new messages to stay updated on communications.
18. As an account holder, I want to exist in multiple organizations, projects and access multiple project-specific chats within each server.
19. As an account holder, I want the ability to delete any messages that I have sent.
20. As an account holder, I would like my documents, files, and chats to be stored securely and protected from SQL injection attacks.

Task Assignee:

21. As a task assignee, I would like to view a list of all the tasks I have been assigned.
22. As a task assignee, I want to view a calendar that shows all the tasks listed on their respective due dates.
23. As a task assignee, I would like to update the completion status of my individual tasks
24. As a task assignee, I would like the ability to notify my task assigner when I make updates to my tasks.
25. As a task assignee, I would like to attach documents to specific tasks that would be shared with the account holder who assigned me the task.
26. As a task assignee, I want to upload documents directly to a task to provide necessary resources or updates.
27. As a task assignee, I want to view and download documents attached to tasks to access shared resources.
28. As a task assignee, I want to receive notifications when new documents are attached to tasks I am assigned to.
29. As a task assignee, I would like to receive email notifications before projects are due.
30. As a task assignee, I would like to be able to select the amount of time before projects are due that I get a reminder notification (i.e. I get an email 3 days beforehand).
31. As a task assignee, I would like to send multiple invitations to users join my project

Task Assigner:

32. As a task assigner, I want to assign different tasks to one or multiple users that allow different levels of permissions to view the task, modify the task or assign the task.
33. As a task assigner, I would like to add or remove people from the task.
34. As a task assigner, I would like to change the permission level (view, modify or assign) of people assigned to the task.
35. As a task assigner, I would like to view a list of all the tasks I have assigned others, with the option to filter by the urgency or status of each project
36. As a task assigner, I would like to assign various levels of importance to tasks.
37. As a task assigner, I want to edit tasks, including the title, text, repository, importance level or deadline.
38. As a task assigner, I want to delete tasks and notify the people assigned to them.
39. As a task assigner, I would like the system to automatically create a group chat that includes everyone assigned to the task if one does not already exist, and to update the permissions of each member as people are added or deleted.
40. As a task assigner, I should be the only one who can add users to my task.
41. As a task assigner, I should be able to make someone else the task assigner if necessary.

Project Manager:

42. As a project manager, I would like to create projects so I can organize my team efficiently.
43. As a project manager, I would like to edit projects so I can organize my team efficiently.
44. As a project manager, I would like to delete projects so I can organize my team efficiently.
45. As a project manager, I want to define project goals and objectives during creation to guide the team.
46. As a project manager, I want to update project details, such as deadlines and descriptions, to reflect in scope.
47. As a project manager, I want to archive completed projects to maintain a clean workspace while preserving historical data.
48. As a project manager, I would like to be able to assign users to a project, with clear communication with the users.
49. As a project manager, I would like the ability to delete any tasks within the project.

Admin:

50. As an admin, I would like to view a list of all users in my organization.
51. As an admin, I would like to manage users' roles so I can control their access to certain features.
52. As an admin, I would like to assign roles such as user, task assigner, project manager, and admin to control feature access
53. As an admin, I would like to be able to add or remove users to my organization.
54. As an admin, I would like to access all private messages between people in my organization, including all deleted messages.
55. As an admin, I want to update user roles as needed to adapt to organizational changes
56. As an admin, I want to notify users of roll changes to ensure they are aware of their new responsibilities
57. As an admin, I want to create an organization

Non-Functional:

Performance

- The system should respond to user actions within one second to ensure a smooth user experience.
- The app should support at least one-hundred simultaneous requests to handle peak usage times.

- The application dashboard should load within five seconds so users can quickly access their projects.
- Task updates should be reflected in less than one second so users can immediately see requests

Scalability

- The system should be able to scale horizontally to accommodate at least 500 user accounts.
- As the system grows, it should not suffer from performance degradation

Usability

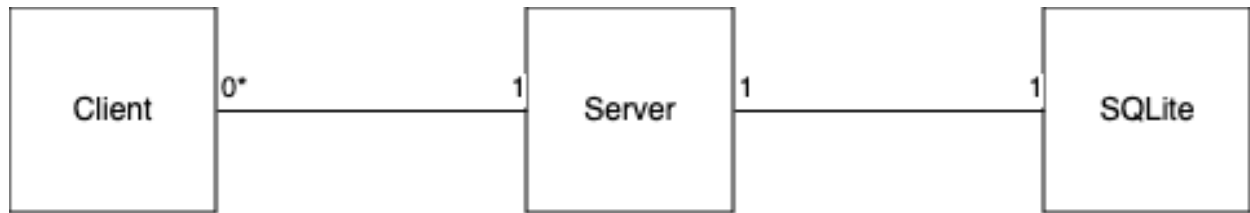
- The app should have an intuitive interface that new users can learn within 30 minutes.
- Navbar: Allows seamless search between all tasks, projects, document repositories, person and chats that you have access to in one search bar.
- Task Tracker Views:
 - Calendar: Visually sorts all tasks based on their due dates. Provides support for weekly, monthly and yearly views, with the ability to look at past and future time periods; based on chronological urgency level
 - To-do list: Lists all tasks with a brief description that allows for users to sort by various attributes, including due date, importance level, task assigner, and date assigned.
 - Kanban Board: Tasks in the form of tiles that move across various columns filtered based on importance level

Security

- User data should be encrypted in transit and at rest to protect privacy.
- The app should implement two-factor authentication for enhanced security.
- User's session should automatically log out after fifteen minutes of inactivity.
- All user actions should be logged to audit activities.
- Data should be filtered before adding to the DB to prevent injection attacks.
- Prevent malicious SQL input from being included in executed queries

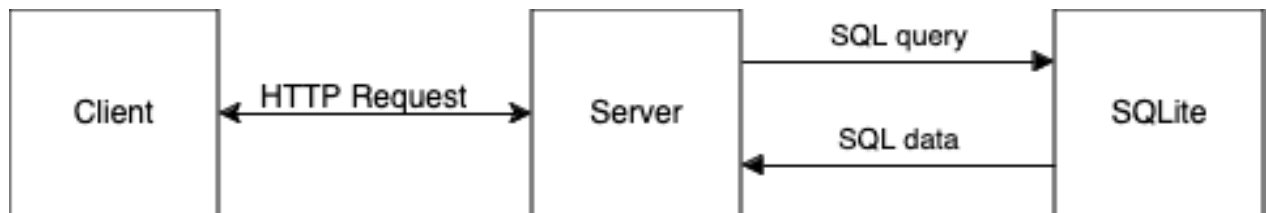
Design Outline

High-Level Overview



- **Client** – The client will handle the users' interaction with the application. It will have multiple views of the project data and be responsive. For our client side, we will be using Vue.js.
- **Server** – The server will handle data transactions between the client and the database. We will be using Spring Boot Java and expose a RESTful API for the client to interact with.
- **Database** – The database will be a SQLite database to store organizational, user, task, and project data.

Interactions Between Components



- The client will send HTTP requests to the server to perform CRUD operations
- The server will listen for incoming HTTP requests, interact with the database through SQL queries, and return data to the frontend

Design Issues:

Functional Issues:

1. What information is required for creating an account?

- A. Username, password, email, phone number
- B. Username, password, email, 2FA info
- C. password, email, phone number, preferred name, 2FA info

Solution: B

We wanted to provide a professional solution that would facilitate easy communication between employees without the confusion of custom usernames. We wanted to use 2FA to provide a secure login system to prevent unauthorized access. We decided against including the user's phone number as we believed it was unnecessary and we wanted to protect the user's phone number from being visible to other people if they don't want it to be public.

2. How should task permissions be handled?

- A. Everyone within a task can make changes
- B. Edit permissions are required to make changes
- C. Everyone within the "parent" project can edit tasks

Solution: B

We want other members of the organization to have the ability to review the progress of a task without being able to edit the document repository and other attributes of the task. Additionally, we only want task assignees to be able to edit tasks. Therefore, "edit" is a specific permission that must be explicitly given, but anyone who is assigned a task can view a task.

3. How are users differentiated within the platform?

- A. With their name and a globally unique user ID
- B. With their name and view email, and their profile photo visible

Solution: B

The user's name will be the primary tool used to distinguish users from a visual perspective, but in large organizations it is possible that multiple users will have the same legal name. The user's email will be included when adding a user, which will necessarily be unique, and their profile photo will also be visible for extra clarity. While more precise, we thought including the user ID would be too confusing for new users.

4. Who can communicate within a task chat?

- A. All members with view access to the task

- B. All members with edit access to the task
- C. All members involved in the parent group
- D. All members involved in the parent organization

Solution: A

We want to facilitate a method of communication so that people in the task can coordinate with each other. Anyone involved in the task can communicate about the task without edit permissions so that everyone is able to contribute. The same goes for project and organization chats.

5. How will chats be organized?

- A. Three levels of chats that can connect users within organizations, tasks and projects
- B. One level of chat that can connect users globally, outside of the structure of organizations

Solution: A

We wanted to organize chats, but also allow for a way for users to communicate with each other as they desired. We decided to create a chat automatically for each organization, project and task, and then let users form group chats within each organization. We decided against letting users directly message other users who do not share an organization with them because we want to empower organizations to moderate their own users, especially in the context of spam and harassment.

6. If the dashboard load time exceeds our 5 second constraint, what will we do?

- A. Keep showing loading bar until we can show it all at once
- B. Iteratively show it as we retrieve dashboard info

Solution: A

The user should never see an incomplete dashboard for a prolonged period of time. We want the user interface to look clean in the general case and not display incomplete information, but we do not want this to dramatically impact the user experience, so once enough time has passed, we chose to simply show the information that has been loaded and show more as it loads in.

Non-Functional:

1. How will security be handled?

- A. None
- B. HTTPS (TLS)
- C. HTTPS and hash passwords

Solution: C

Implement TLS security between the client and the server. Do not store passwords in plaintext, but only store a hash. Implementing these features will prevent man-in-the-middle attacks.

2. What Front-end framework will be used?

- A. Vue.JS
- B. React.JS
- C. Angular

Solution: A

Vue.js is ideal for beginners due to its easy learning curve and comprehensive documentation. It offers flexibility for projects of any size, a component-based architecture for maintainability, and reactive data binding for seamless UI updates. With strong community support and a lightweight, performant nature, Vue.js is a progressive framework that can be adopted incrementally, making it a great choice for new front-end developers.

3. What database will be used?

- A. PostgreSQL
- B. SQLite
- C. MySQL
- D. MongoDB

Solution: B

We chose SQLite for its ease of use when making lightweight applications. It requires minimal configuration, allowing quick setup and use. SQLite is optimized for read-heavy and low-concurrency applications, so it is good for a low traffic site like Aegis. We prioritized ease of deployment and minimal maintenance, making SQLite ideal for our application.

4. Will we have global projects and global task tables?

- A. One large table for each category (i.e. one project table, one task table, etc.)

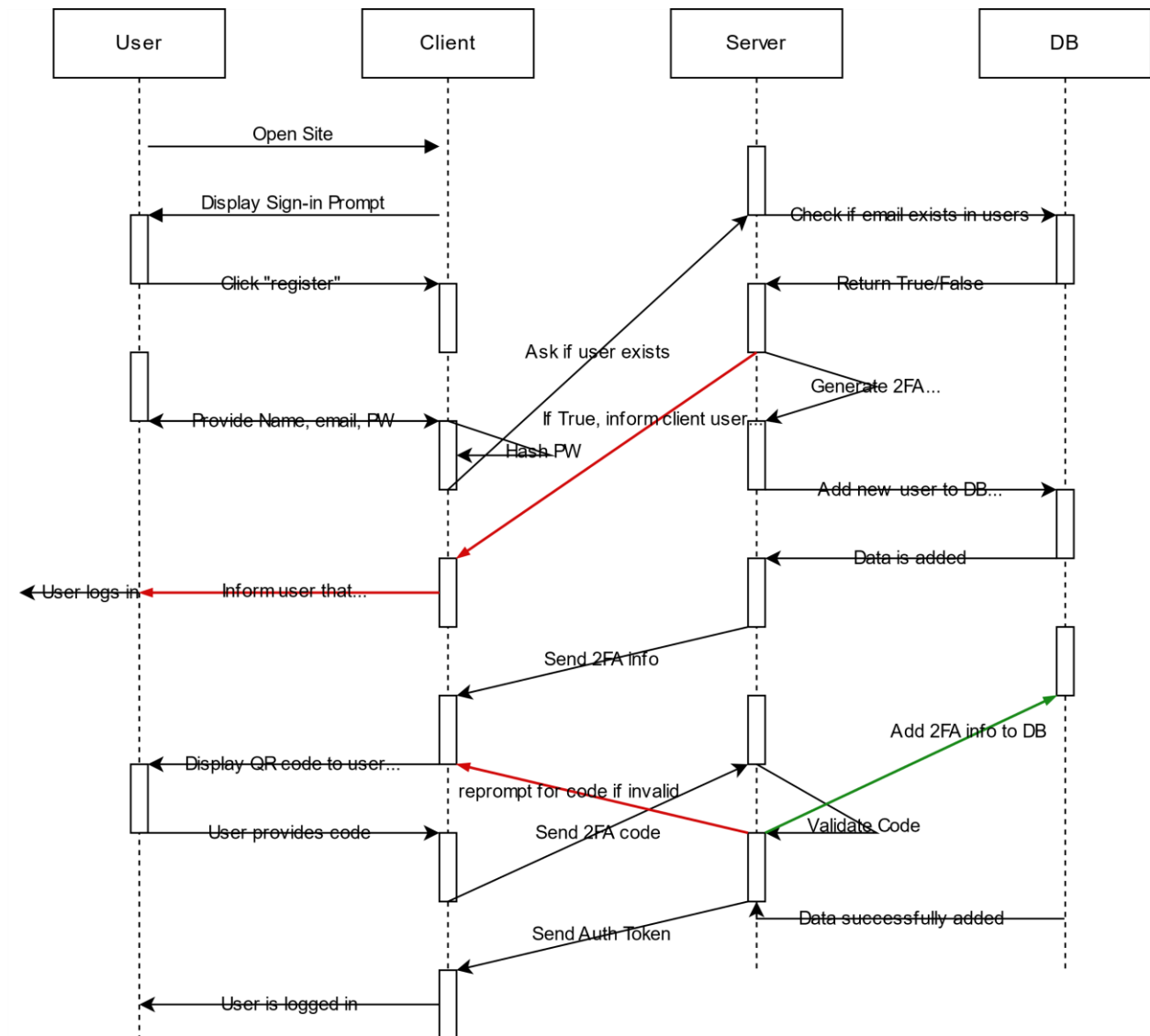
- B. A distinct table for each subset in the hierarchy (i.e. a unique project table for all projects belonging to a specific organization).

Solution: B

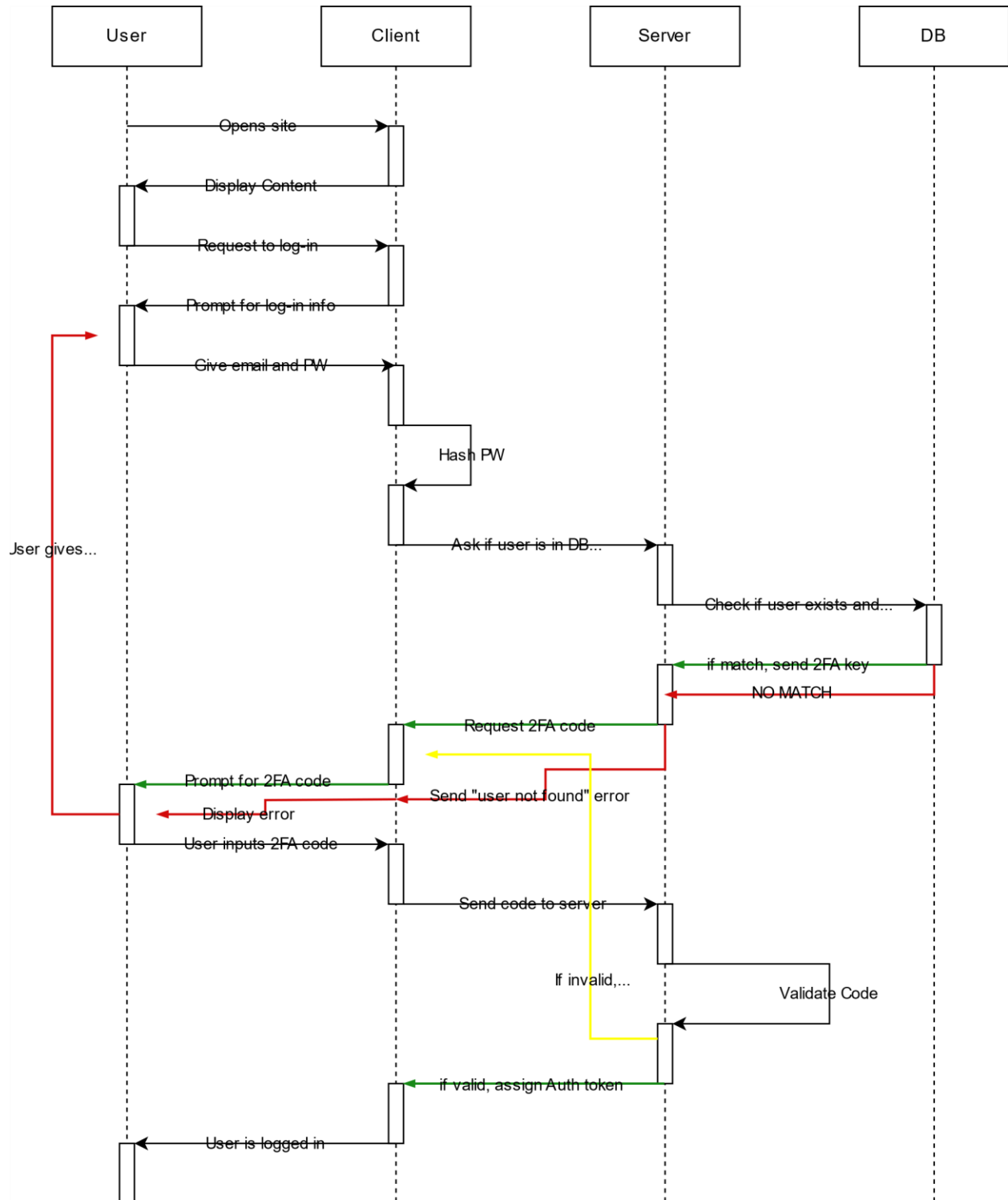
We wanted a way to organize tables while keeping the design simple. For each organization, they will have their own table of projects, and each project will have their own table of tasks. We wanted to have an organized database that would avoid repeated data and would result in a cleaner code base.

Sequence Diagrams:

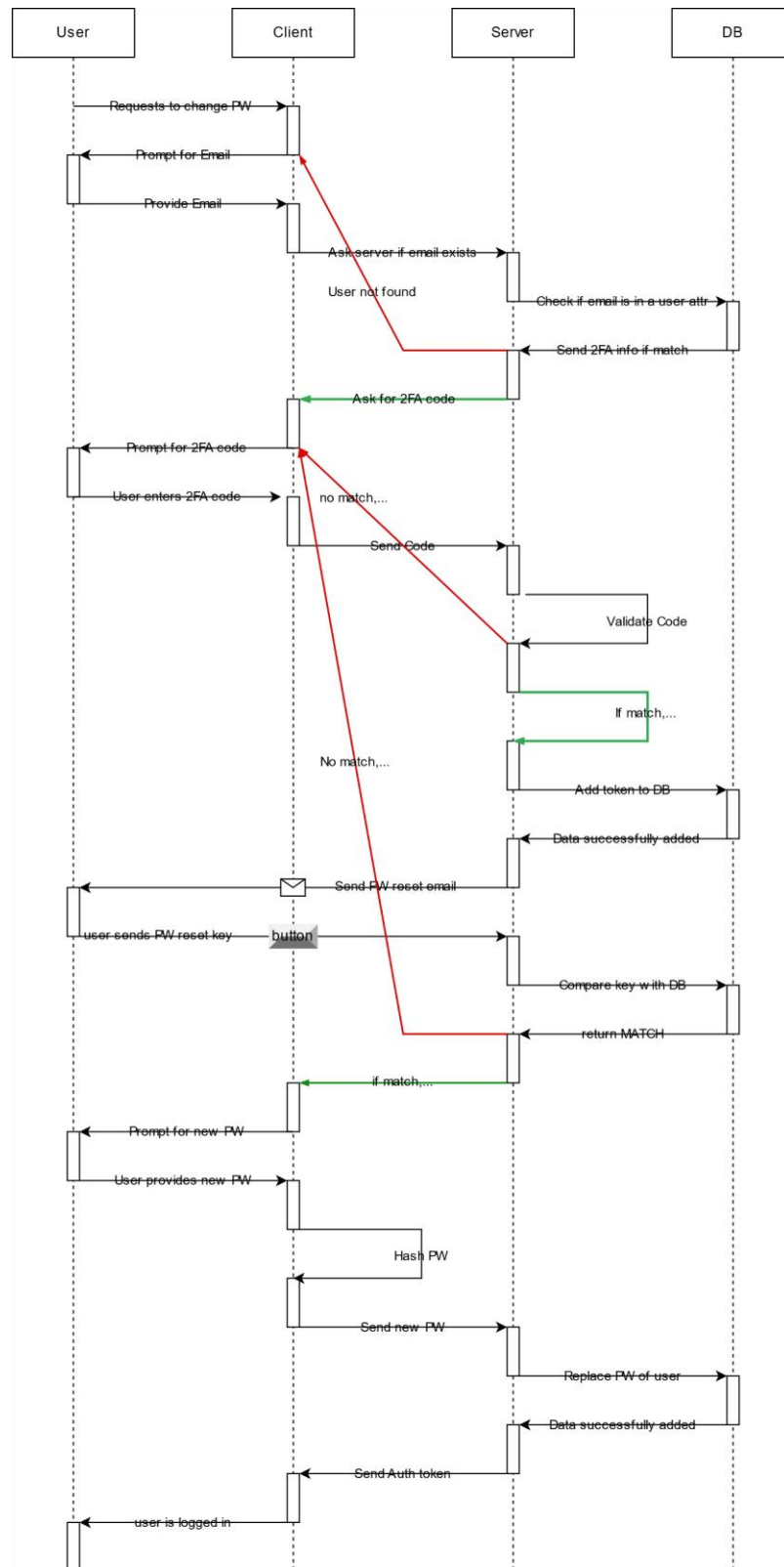
Create Account:



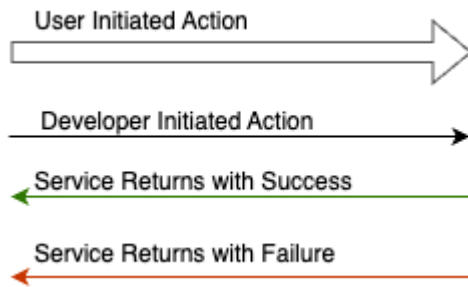
Log in



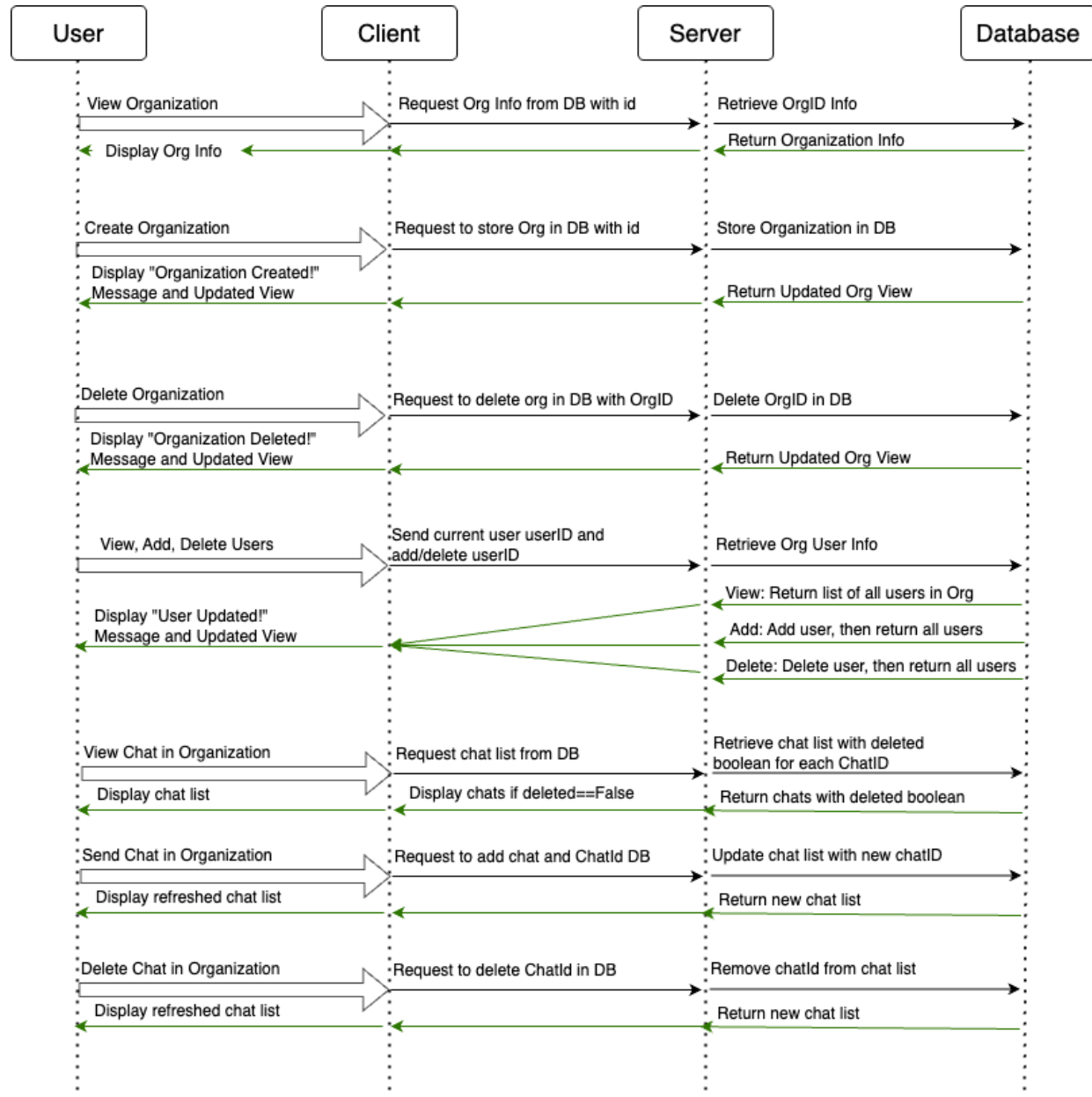
Change password



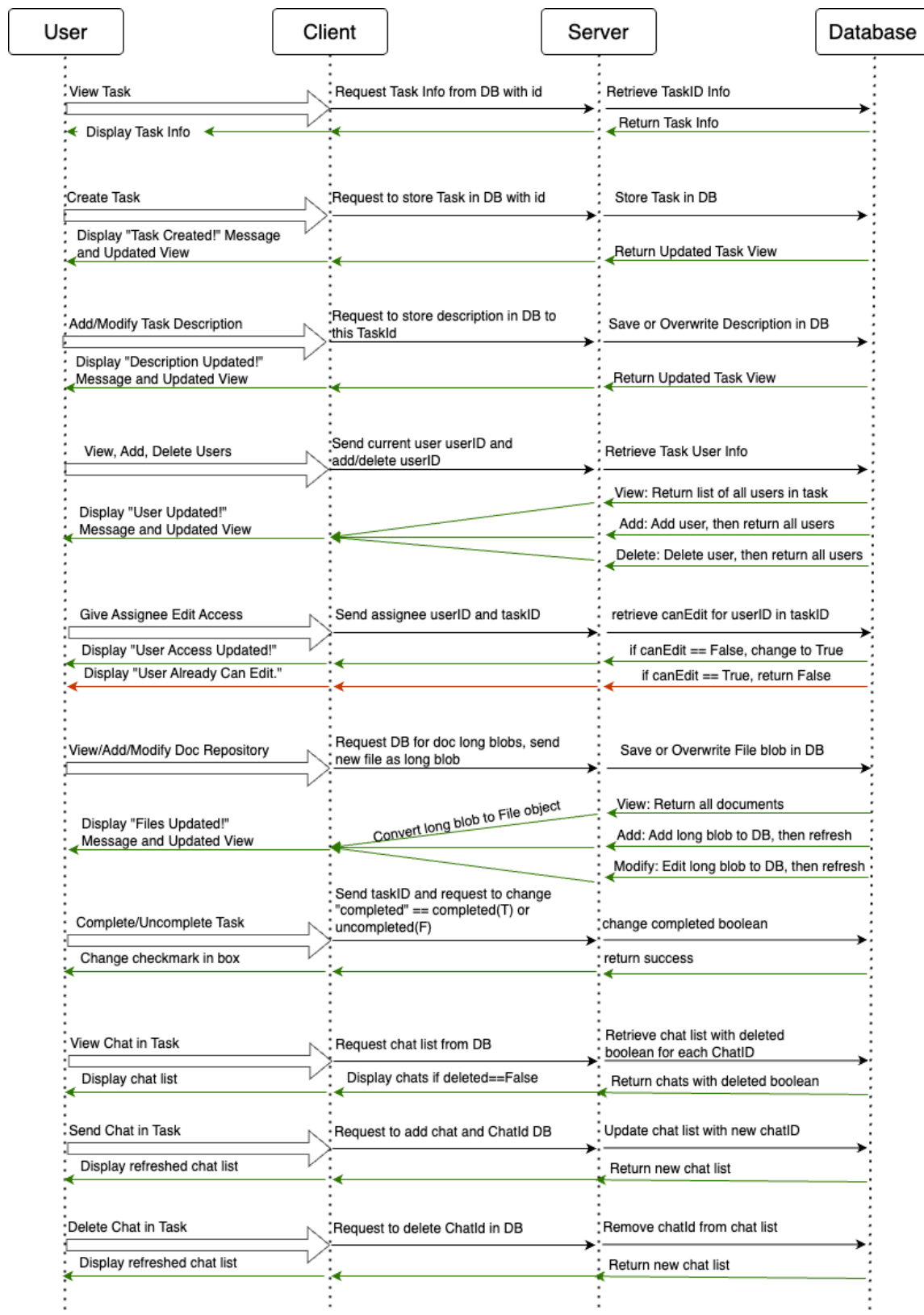
Organization Related Actions



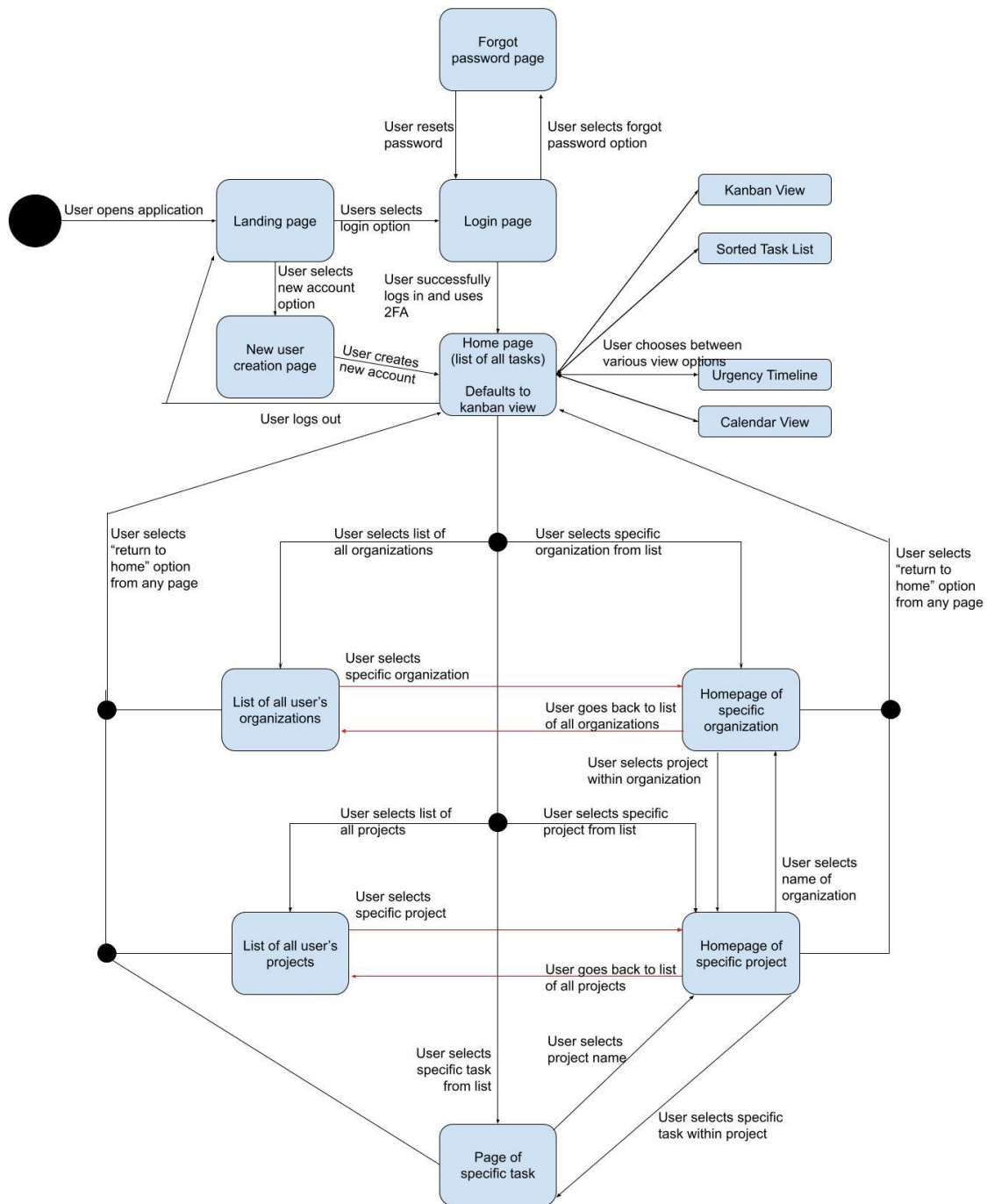
Note: the **next two diagrams** define arrows as per this legend:



Task Related Actions



Navigation Flowchart for User Interface

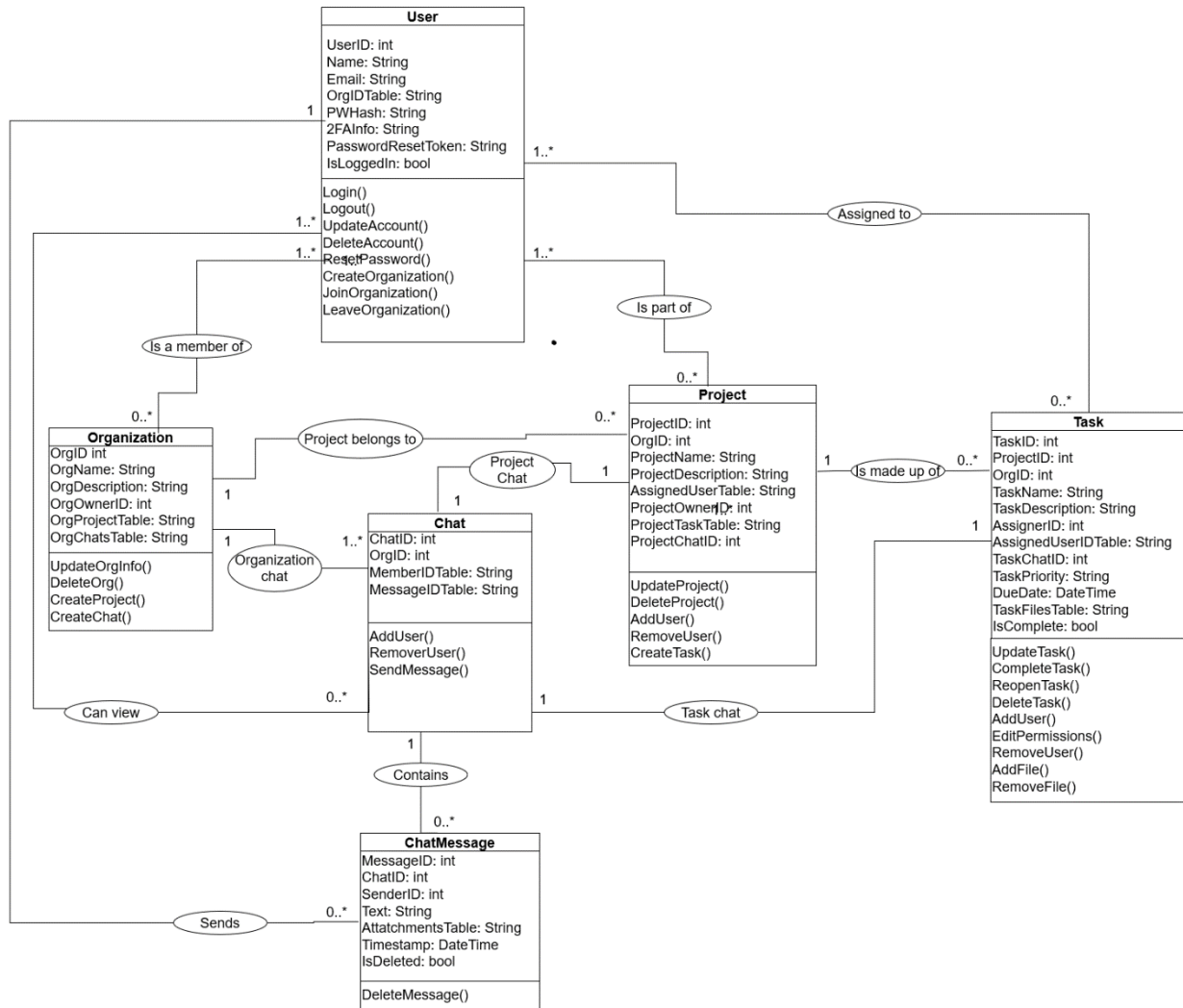


- When the user launches the website, they are directed to a landing page that greets them, includes an about me section talking about the purpose of the application, and are prompted to either create a new account or log into an existing account
 - If they choose to create a new account, they are given the ability to input their name, email, password, etc. into a short form on a new page, which will verify that the email is unique and then create a new account.
 - If they choose to log into an existing account, they are given the ability to enter their email and password. If they have forgotten their password, they are directed to a new page where they will be given the ability to enter a code sent to their email. If the code matches, they will be prompted to create a new password and will then be redirected back to the login page.
- Once they have successfully logged in, they will be greeted by their home page, which has a list of all their tasks across all the different projects and organizations that they are involved in. They can view this tab in several different ways:
 - Kanban view (default): Groups the user's tasks into 3 columns based on whether they have not been started, whether they are in progress, or whether they have been recently finished.
 - Sorted task list view: Lists all the users' tasks in one table, with their description, priority, deadline and completion status listed. Includes a search bar at the top to search for tasks or filter based on specific attributes
 - Urgency timeline view: Groups tasks into different categories by due date and urgency status, highlighting which ones are almost due and which ones are most important
 - Calendar view: Shows all tasks sorted by due date into a monthly calendar view, enabling the user to visualize which tasks are due when
- They are also able to travel to different pages from the home page to focus in on their organizations, projects and specific tasks
 - They can view a list of all of the organizations they are a part of, sorted by most recent activity. From there, they can click on any specific organization to view that organization's page, and can travel back to the complete organization page
 - They can also view a list of the projects they are involved in, sorted by most recent activity. They can similarly click on a specific project to view the project's page and travel back as well.
 - Their most recently updated organizations and projects are directly accessible from their home page as well, which can be selected to bring them directly to the specific organization/project page.
 - The user can click on any of their tasks, either from the list on their home page or the list of tasks on their project page, to view the status of the task and update anything on their end if they have the permission to do so.
 - The user can access their chats and return to the home page from any of these tabs

- Finally, the user can log out of their account from their home page, at which point they will be redirected to the landing page and can either leave the website or log into a different account.

Design Details

Class Design



User

- The User class represents each individual account that is created.
- Every User has a global ID that uniquely identifies them in a globally accessible table.
- Each User consists of their name, email address, a hash that represents their password in a securely stored fashion, and a String that represents their code for two-factor authentication. It also creates a token to validate the user's identity in the event they forget their password.
- It also contains a String called OrgIDTable that represents the name of a table that contains the OrgIDs of all the organizations that the user belongs to (stored as ints)
- The User can update the attributes of their account and delete their account
- The User can also create a new organization, join an existing organization with the ID and leave an organization that they belong to.

Organization

- The Organization class represents organizations as a whole and can be created by a User
- Every Organization has a unique global identifier
- Each Organization consists of its name, description, the User ID of its owner, a table for tracking its projects, and the Chat IDs for both the main, Organization wide chat and any private chats within the Organization
- The owner can edit the attributes of the Organization and delete it altogether
- The Organization manages the creation of new projects and chats

Project

- Each Project exists under the umbrella of an Organization, representing a specific group of people that can assign Tasks to one another.
- Projects are uniquely identified with an int ID called ProjectID that corresponds to its position in the table of all Projects that belong to its specific Organization. It also stores the ID of the project it belongs to in OrgID.
- The Project has a name, text description, the name of a table that contains the IDs of all users that have access to the project, the ID of the specific user that created the project (who is the only one with the ability to change the name or description, or add or remove people from the project)
- The Project also has the name of the table that stores all the Tasks associated with that Project, as well as the ID number of the Chat that corresponds to the specific Project.
- The Project can have its information updated, the project itself deleted, users added or removed, and new Tasks created.

Task

- The Task class represents a task that must be completed as part of a project

- Each Task exists within the constraints of an associated “parent” Project
- Every Task has an identifier that is unique on a Project level
- Each Task consists of a name, description, the IDs for its “parent” Project and Organization, the User ID of the task assigner, a table for tracking task assignees and their permission, the Chat ID for the task wide chat, task priority level, due date, a table for tracking task files, and a value to check completion
- The assigner of the Task can edit its attributes, delete the Task, add or remove assignees, and edit the permissions of the assignees
- Any task member with edit permissions can add or remove files and mark the task as complete or incomplete

Chat

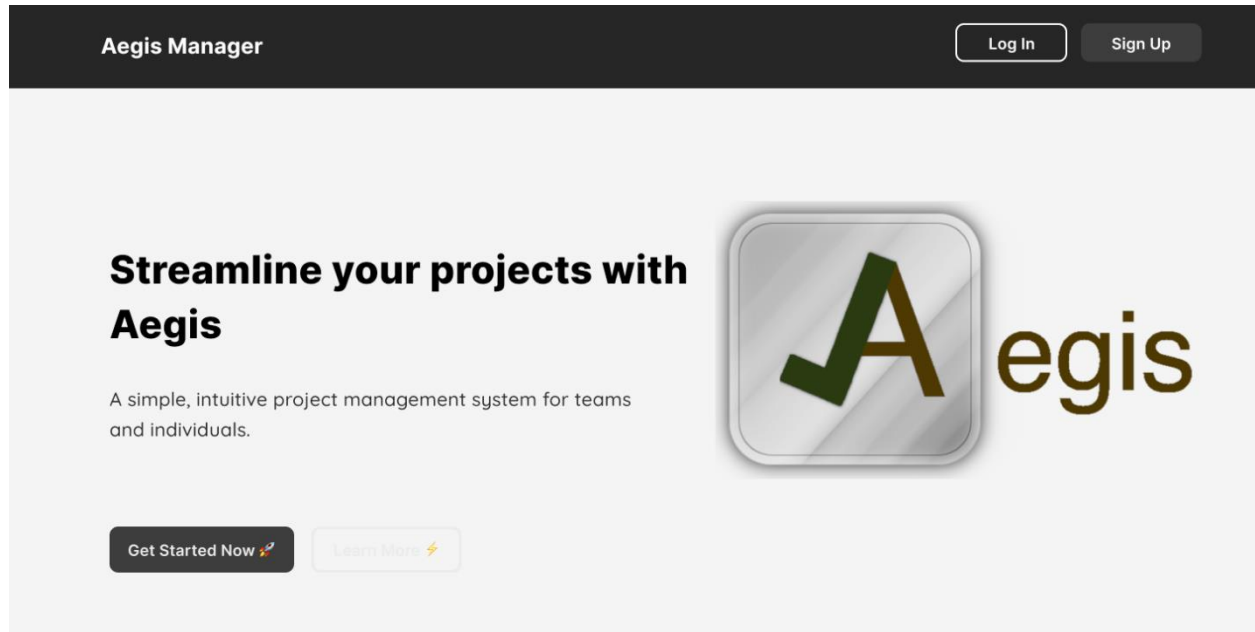
- A Chat consists of a group of people that belong to a shared Organization who can send direct messages and files to one another, either on a one-to-one basis or in larger groups.
- Each Chat is part of an Organization (only people who share an Organization can chat with each other) and stores the ID of the Organization that it belongs to.
- A Chat is automatically created when an Organization, Project or Task is created that will contain each group's (catch-all for Organization, Project or Task) members and add or remove people at the same time as they are added or removed from the group.
- Users can also form Chats with other members of any Organizations they belong to independently of Groups or Projects
- Chats will store the name of a table that contains all the ChatMessages sent in that Chat, as well as the name of the table that contains all the Users that have access to the Chat.
- Chats will contain functionality to send ChatMessages

ChatMessage

- A ChatMessage represents each individual message that a User sends in a Chat.
- Each ChatMessage has an ID that uniquely identifies it in the table that consists of all the ChatMessages sent within a specific Chat.
- ChatMessages also store the ID of the person who sent it, the text of the message, the name of a table that contains all the file attachments for the message, the timestamp at which the message was sent
- ChatMessages contain functionality to delete a message, and a bool representing whether the message has been deleted (it will still exist and be visible to the organization owner, but it will not be visible to the sender or recipients)

UI Mock-Ups

Landing Page



Account Creation & Sign-In

Sign Up

Log In

Sign Up

Create Profile

Register Securely with Aegis

1

General Information

First Name

Last Name

Email Address

City/State

Profile Picture

Upload Image



Or Sign In

Register Securely with Aegis

1

General Information

Email

Password

Upload Image

View of All Tasks

Ridgeline UI Design

Your Tasks

Search

Filter

Deadline

Task	Description	Priority	Due Date	Completed?
Design Social Media Assets	Design a set of elements...	High	2023-03-15	☒
Create Email Marketing Template	Design and develop an email marketing template...	Medium	2023-03-18	☒
Plan Event Logistics	Coordinate event logistics for the new product...	High	2023-03-20	☐
Write Press Release	Write a press release announcing the launch...	Medium	2023-03-22	☐
Develop Landing Page	Design and develop a landing page	High	2023-03-25	☐
Conduct Market Research	Conduct market research to identify target audience...	Low	2023-03-28	☐

User's Kanban Page (Alternate View of All Tasks)

Aegis

Sophia James

soph_james@SC2.com

Product Owner & Designer

List View

To Do3

Conduct Market Research

Mar 28

Develop Landing Page

Mar 25

Plan Event Logistics

Mar 20

In Progress1

Write Press Release

May 22

Done2

Create Email Marketing Template

Mar 18

Design Social Media Assets

Mar 15

User's Projects

SC2 Data Refresh

Ridgeline UI Design

Bing Merger Docs

Project View

Aegis



All Projects

Search All Projects

Task

User

Deadline



SC2 Data Refresh
Blockchain Division



3 tasks pending
14 unread chats



Ridgeline UI Design
UI/UX Division



1 tasks pending
23 unread chats



Bing Merger Docs
Blockchain Division



12 tasks pending
3 unread chats



SC4 Pipeline and Bing Integration
SWE Division



0 tasks pending
140 unread chats

Individual Project Page

Project

Chat

Settings

Project

Ridgeline UI Design

Created by

In Progress

09/29/2024

John Doe

Project Description

This project aims to plan and execute a comprehensive UI campaign for the launch of our new product, including social media, email marketing, and event planning.

Attached Documents

Photos Library...toslibrary

Amarr Tempest 8k.png

7680x4320

6348844560.webm

wp5539411.webp

5120x2880

A

Task 1

☒

A

Task 2

☒

A

Task 3

☒

A

Task 4

☒

A

Task 5

☒

A

Task 6

☒

A

Task 7

☒

A

Task 8

☒

A

Task 9

☒

A

Task 10

☒

View all Tasks

Chat

No messages yet

New Message

View Individual Task:

TaskChat

Task 7:
Design Social Media Assets

High09/17/24Over Due!

Completed this Task?
☒

About this task
Design a set of social media assets (Facebook, Twitter, Instagram, LinkedIn) for the new product launch campaign, including banners, posts, and profile pictures

Assigned to you by:
Trevor Smith

Attached Documents

Previous

Photos Library...toslibrary

Amarr Tempest 8k.png
7,680 x 4,320

6348844560.webm

wp5539411.webp
5,120 x 2,880

Chat

No messages yet

New Message

Organization Level Chat

CoFounder Match

MatchInboxSettings

Kevin Smalls
Can you bring me a ladder?

Mr. Weed
Got some new products in...

G-Man
Rise and Shine Dr. Freeman!

Kevin Smalls

Can you bring me a ladder?

Sure thing, buddy!